

Vector databases: The Backbone of AI-Powered Search and Retrieval

Introduction

- ▶ Traditional databases store structured/tabular data.
- ▶ AI applications (e.g., search, recommendation) need to work with unstructured data like images, text, and audio.
- ▶ Vector databases store and search **embeddings** — numerical representations of unstructured data.

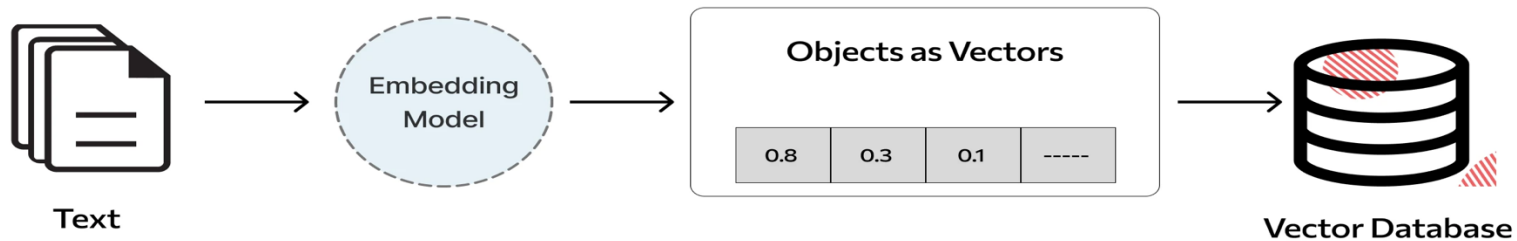
What Is a Vector?

- A vector is an array of floating-point numbers.
- It represents the **semantic meaning** of data.
- Generated by ML models like BERT, CLIP, or OpenAI Embeddings.

Example: "dog" $\rightarrow$ [0.12, -0.32, ..., 0.87]
"puppy" will be close to "dog" in vector space.

What is vectordb?

Vectordb is a data bases that stores embeddings of words, phrases and documents along with corresponding identifiers



Why We Need Vector Databases

- ▶ Unstructured data is 80% of data.
- ▶ Vector databases are really good for unstructured data.
- ▶ Examples of unstructured data include text, audio, images, etc.

Why We Need Vector Databases

Challenge	Traditional DB	Vector DB
Full-text search	✓	✓
Semantic similarity search	✗	✓
Image/audio search	✗	✓
Neural search	✗	✓

How Vector Search Works

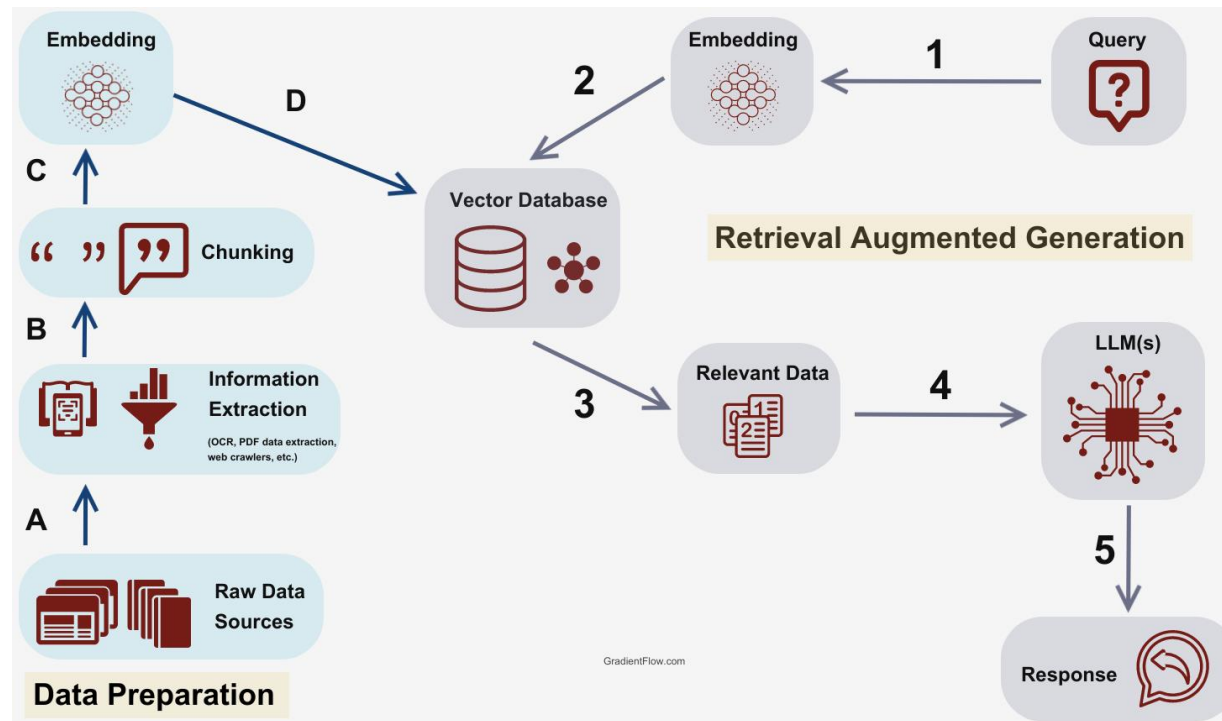
Steps:

1. Convert data to vectors using embedding models.
 2. Store vectors in the database.
 3. Search using a **query vector** and retrieve **nearest neighbors**.
- **Core Technique:** Approximate Nearest Neighbors (ANN)
– Uses HNSW, IVF, PQ, etc. for efficient search.

Common AI Use

-  **RAG combines LLMs with external knowledge** by retrieving relevant documents from a vector database before generating a response.
-  It reduces hallucinations by grounding answers in **factual, context-rich sources** like PDFs, wikis, or internal docs.
-  Enables **domain-specific Q&A** without fine-tuning the LLM, making it faster and cheaper to deploy.
-  Uses **embedding models + vector search** to find semantically similar chunks of data at query time.
- ▶  Powers applications like chatbots, knowledge assistants, legal search, and healthcare copilots.

Architecture Overview



Applications

- ▶ Knowledge management and information retrieval
- ▶ Product recommendations
- ▶ Customer support and chatbot
- ▶ Content marketing
- ▶ Research and Development
- ▶ Fraud detection